STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Recap: One Target, Four Approaches

g-computation and matching

g-computation: model outcomes under $T = 1$ and $T = 0$.

Matching: compare observed units with similar baseline covariates.

IPW and AIPW

IPW: weight by treatment probabilities.

$AIPW$: combine an outcome model and a propensity model.

Each method needs a credible identification strategy.

Recap: Four Estimation Strategies

g-computation

Predict both potential-outcome means and average their difference.

IPW

Reweight observed outcomes to create a pseudo-population.

AIPW

Combine predictions with inverse-weighted residual corrections.

AIPW is doubly robust to either nuisance model being correctly specified, not to unmeasured confounding.

Recap: Identification versus Estimation

Identification

Consistency, exchangeability, positivity and appropriate interference assumptions connect data to causal effects.

Estimation

Regression, matching, weighting, doubly robust estimators and forests estimate effects after identification.

More flexible prediction cannot, by itself, cure unmeasured confounding.

Recap: ATE and CATE

$$ATE = E[Y(1) - Y(0)] \quad \text{vs.} \quad CATE(x) = E[Y(1) - Y(0) \mid X = x]$$

ATE asks: what is the average effect in the population?

CATE asks: how does the average effect vary across baseline characteristics X?

CATE is a subgroup conditional mean effect, not an observed individual treatment effect.

Recap: Interaction Models

$$E[Y \mid T, X] = \beta_0 + \beta_1 T + \beta_2 X + \beta_3 (T \times X)$$

The treatment effect at $X = x$ is $\beta_1 + \beta_3 x$.

β_3 describes how the treatment effect varies with X.

Centering X changes the meaning of β_1, not model predictions.

Recap: Three Meta-Learners

S-learner

Fit one outcome model with T and X; predict two treatment worlds.

T-learner

Fit separate outcome models for treated and controls; subtract predictions.

X-learner

Impute treatment-effect signals, fit effect models and combine them.

These are modeling strategies, not different causal identification assumptions.

Recap: S-Learner and T-Learner

S-learner

Use the full sample in one model $m(T, X)$.

CATE estimate: $m(1, x) - m(0, x)$.

Risk: treatment effect may be shrunk toward zero.

T-learner

Fit $m_1(x)$ among $T = 1$ and $m_0(x)$ among $T = 0$.

CATE estimate: $m_1(x) - m_0(x)$.

Risk: small treatment arms can give unstable fits.

Both can use linear models, GAMs, random forests or other outcome learners.

Recap: Why the X-Learner Is Different

First stage: outcome models

Estimate $m_1(x)$ and $m_0(x)$.

For treated: $D^1 = Y - m_0(X)$.

For controls: $D^0 = m_1(X) - Y$.

Second stage: effect models

Regress D^1 and D^0 on X separately.

Combine fitted effects using propensity-based weights.

Can be useful under treatment imbalance.

Pseudo-effects are constructed outcomes, not individually observed causal effects.

Recap: Causal Forest versus Ordinary Forest

Ordinary random forest

Build splits that improve prediction of Y.

Excellent outcome prediction need not reveal treatment-effect differences.

Causal forest

Build splits aimed at treatment-effect heterogeneity.

Estimate local treatment effects using nearby observations.

Recap: Honesty and Cross-Fitting

Honest forests

Use different subsamples to select splits and estimate effects.

Reduce adaptive overfitting and estimation bias.

Cross-fitting

Fit nuisance models on other folds.

Generate out-of-fold predictions for the evaluation observations.

Sample splitting helps estimation validity; it does not remove unmeasured confounding.

Recap: How to Evaluate HTE

Calibration

Do larger predicted effects correspond to larger observed causal contrasts?

Subgroup comparisons

Compare groups in independent data using valid causal estimators.

Uncertainty

Report intervals, sampling variability and multiple-testing concerns.

Variation in predicted CATEs alone is not proof of real treatment-effect heterogeneity.

STAT 7900: Quiz 8



Recap: The Question We Face Today

What if an important confounder U was never measured?

Matching, IPW and AIPW adjust for observed X.

HTE learners and causal forests estimate effects conditional on X.

None automatically eliminates bias from an unobserved common cause U.

Today: sensitivity analysis quantifies fragility; instrumental variables use a different design.

A Concrete Reason to Worry: Missing Motivation

Example: job-training program

Some participants enroll because they are especially motivated.

Motivation may also improve their earnings even without training.

What the dataset records

Treatment, age, education, prior earnings, and later earnings.

But motivation was not recorded. Participants and nonparticipants can still differ.

Discussion: if we exactly balance age and prior earnings, is the comparison necessarily causal?

Estimation Is Not Identification

Estimation problem

Given an identified causal effect, how can we estimate it reliably?

Regression, IPW, AIPW, and forests address aspects of this problem.

Identification problem

Can observed outcomes reveal the causal effect at all?

If U affects both T and Y, adjusting only for measured X may not identify the effect.

A more powerful prediction algorithm cannot reconstruct an unobserved confounder automatically.

Challenge the Assumption

Question 1: What could go wrong?

An unmeasured common cause U makes treated and control groups incomparable even after adjustment.

We can examine its causal diagram and possible bias.

Question 2: How fragile is the conclusion?

Instead of assuming U has no influence, assess how strong hidden confounding would need to be to overturn a result.

Tools: E-values and Rosenbaum bounds.

Sensitivity analysis tests robustness to specified hidden bias; it does not eliminate the bias.

When the Ignorability Assumption Fails

What we have assumed

$$Y(0), Y(1) \perp T \mid X$$

After conditioning on measured X, treatment assignment is unrelated to potential outcomes.

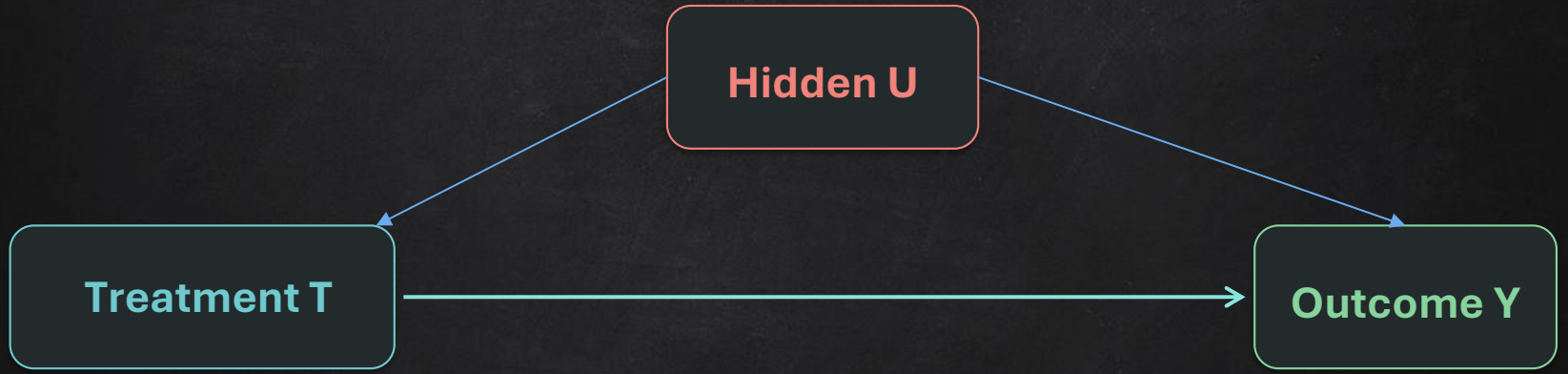
The difficult possibility

A variable U influences treatment and the outcome, but U was never recorded.

Conditioning on X may no longer remove confounding.

More sophisticated estimation does not automatically repair failed identification.

A Hidden Common Cause: The DAG



$U \rightarrow T$: influences assignment

$U \rightarrow Y$: predicts outcome

Open backdoor path: $T \leftarrow U \rightarrow Y$

Example: Treatment Selection by Severity

Clinical setting

Patients with more severe illness are more likely to receive an aggressive treatment.

Severity also increases the probability of a poor outcome.

What goes wrong

Suppose the recorded data include age and sex, but not severity.

Comparing treated and untreated patients can make an effective therapy appear harmful.

The apparent treatment association mixes treatment effects with selection effects.

Why Familiar Estimators Cannot Fix Hidden U

Matching / IPW

Balance measured X; U can remain imbalanced.

g-computation / AIPW

Model outcome and treatment using measured X, not unobserved U.

S/T/X-learners

Flexible response models do not guarantee exchangeability.

Causal forests

Better CATE estimation does not identify effects without a valid design.

Bias from unmeasured confounding is an identification issue, not merely prediction error.

Numerical Illustration: A Spurious Effect

True causal effect $= 0$

Suppose treatment has no effect.

Among treated: 80% are high risk.
Among controls: 20% are high risk.

Risk of outcome: high 30%, low 5%.

Observed comparison

Treated risk = $.80(.30) + .20(.05) = .25$

Control risk = $.20(.30) + .80(.05) = .10$

Observed risk difference = $+0.15$

A 15-percentage-point observed gap can exist even when treatment changes nothing.

What Is Sensitivity Analysis?

Conventional analysis

Estimate the treatment effect under assumptions about measured confounders and identification.

Sensitivity analysis

Vary assumptions about unmeasured confounding; examine how much conclusions would change.

It tests robustness to specified departures from assumptions; it does not verify them.

The Key Question: How Much Hidden Bias?

Weak U

Small imbalance and weak outcome association.

Conclusion may persist.

Moderate U

Can attenuate the estimated effect.

Intervals may include null.

Strong U

May overturn the estimated sign or statistical conclusion.

A tipping point is the strength of U at which the conclusion changes.

Different Sensitivity Tools Answer Different Questions

E-values

How strongly must U relate to exposure and outcome (RR scale)?

Rosenbaum bounds

How much hidden treatment-assignment bias can matched comparisons tolerate?

Quantitative bias analysis

Choose confounder prevalence and outcome associations; compute corrected estimates.

Alternative identification

Negative controls, IVs, and designs use additional assumptions and information.

E-Values: Intuition

A robustness summary on the risk-ratio scale

For an observed exposure–outcome risk ratio, the E-value summarizes the minimum association strength an unmeasured confounder would need with BOTH exposure and outcome, conditional on measured X, to explain away the association.

Larger E-values generally indicate greater robustness to this specific kind of confounding.

E-values do NOT measure how likely the hidden confounder is to exist.

E-Value Formula: Risk Ratio Above 1

For $RR \geq 1$ (Risk Ratio (RR) – Risk in treatment group / Risk in Control Group)

E-value(RR) = $RR + \sqrt{[RR \times (RR - 1)]}$

For a confidence interval entirely above 1, compute a second E-value from the lower confidence limit.

If the interval includes 1, the interval-limit E-value is 1.

Use the RR scale. OR / HR and continuous outcomes require appropriate care or conversion.

Worked E-Value Example

Suppose a study reports

Adjusted risk ratio = 2.00

95% confidence interval: [1.30, 3.00]

Outcome: a binary adverse event.

Calculate

Point: $2 + \sqrt{[2(2-1)]} = 3.41$

CI limit: $1.3 + \sqrt{[1.3(.3)]} = 1.92$

E-values: point 3.41; CI 1.92

The CI-bound E-value is often more relevant to whether the result remains statistically distinguishable from null.

R Code: Compute E-Values

```r
evalue_rr <- function(rr) {
  r <- ifelse(rr < 1, 1 / rr, rr)
  r + sqrt(r * (r - 1))
}

rr <- 2.00
ci <- c(1.30, 3.00)
evalue_rr(rr)       # 3.414
evalue_rr(ci[1])    # 1.924
```

Important

This code illustrates risk ratios and a CI above 1.

For ratios below 1, use the reciprocal and the CI limit nearest 1.

Not a substitute for judging U's plausibility.

What E-Values Do, and Do Not Establish

Useful

A compact numerical benchmark.

Can compare sensitivity against measured covariate associations.

Simple to calculate and report.

Limitations

Cannot prove no confounding.

Does not establish the existence of a specific U.

Does not repair measurement error, selection bias, or model/design flaws.

Wrong to say: “an E-value of 3.4 means the causal effect is proven.”

Rosenbaum Bounds: Motivation

After matching

Two patients appear comparable on observed X.

But one may still have a higher probability of receiving treatment because of unmeasured U.

Rosenbaum sensitivity analysis

Ask how large the hidden difference in treatment odds could be before a matched-study significance conclusion becomes fragile.

Rosenbaum bounds focus on sensitivity of matched comparisons to hidden assignment bias.

Rosenbaum's Γ: Hidden Assignment Bias

Interpretation of Γ

$\Gamma = 1$: No hidden differential odds of treatment within a matched pair.

$\Gamma = 1.5$: Two matched individuals could differ in treatment odds by up to a factor of 1.5 because of unmeasured factors.

$\Gamma = 2$: Treatment odds could differ by up to a factor of 2.

Γ is an odds-ratio bound for assignment, not an outcome risk ratio or bias percentage.

Worked Example: Matched Pairs and Γ

Imagine 10 matched pairs

Each pair has similar measured baseline characteristics.

Treatment group often has higher outcome ranks than matched controls.

At $\Gamma = 1$, a paired Wilcoxon test may reject the null.

Increase hidden bias

At $\Gamma = 1.25$, assess worst-case p-value bounds.

Repeat for $\Gamma = 1.5, 1.75, 2$ ...

The smallest Γ at which the upper p-bound exceeds .05 marks loss of worst-case significance.

Illustrative procedure; the tipping Γ must be computed from the actual matched outcomes.

R Code: Rosenbaum Bounds for Matched Pairs

```
# install.packages("rbounds")
library(rbounds)

# Illustrative paired outcomes
trt  <- c(12, 15, 11, 20, 18, 19,
          16, 14, 13, 17)
ctrl <- c( 8, 10,  9, 12, 13, 14,
          11, 10, 12, 13)

psens(trt, ctrl, Gamma = 2,
      GammaInc = 0.25)
```

How to read output

Check upper and lower p-value bounds over Γ.

If upper bound becomes > .05, worst-case statistical significance is no longer guaranteed.

Input vectors must be correctly paired.

Matched Pairs: Hidden-Bias Sensitivity in Numbers

Γ (assignment odds bound)	Worst-case upper p-value
1.00	0.0010
1.25	0.0028
1.50	0.0060
1.75	0.0109
2.00	0.0173
2.50	0.0346
3.00	0.0563

Exact upper p-bound for a one-sided sign test with 10 positive pair differences.

Putting the Results Into an Applied Report

What to report

Effect estimand, estimate, and uncertainty.

Measured adjustment set and balance diagnostics.

Chosen sensitivity method and assumptions.

Numerical sensitivity parameter(s).

What to conclude

Whether conclusions are robust within a stated range.

What plausible hidden confounders might look like.

Where conclusions change.

Avoid claims that the analysis rules out all bias.

Sensitivity analysis supplements, rather than replaces, research-design credibility.

Summary and Transition to Instrumental Variables

Takeaway message

Unobserved U can invalidate exchangeability.

Sensitivity analysis examines robustness to specified hidden confounding.

E-values and Rosenbaum bounds answer different questions.

Next: Instrumental Variables

Can we identify an effect when exchangeability fails?

We will need new assumptions: relevance, exclusion, independence, and often monotonicity for LATE.

Sensitivity: “How fragile?” | IV: “Can a new design identify an effect?”

From Sensitivity Analysis to Instrumental Variables

Sensitivity analysis

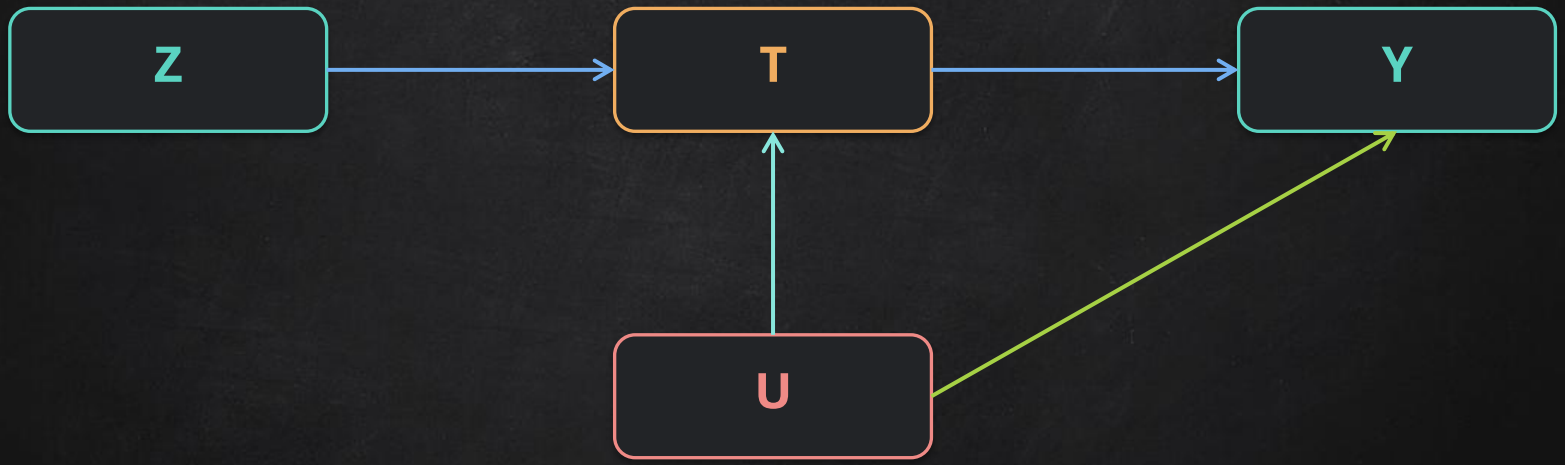
How strong must hidden confounding be to change our conclusion?

Instrumental variables

Can extra design information identify a causal effect despite hidden confounding?

Sensitivity describes fragility; IV changes the identifying assumptions.

Instrumental Variables: The Causal Structure



Relevance: $Z \rightarrow T$ | Exclusion: no direct $Z \rightarrow Y$ | Independence: no $Z - U$ *association*

A valid instrument moves treatment without an independent route to the outcome.

Three Core IV Assumptions

1. Relevance

Z predicts T, even after conditioning on any specified baseline X.

2. Independence

Z is independent of the potential outcomes and relevant unmeasured confounders.

3. Exclusion

Z changes Y only through its effect on treatment T.

The assumptions are substantive design claims, not guaranteed by a fitted model.

IV Assumptions: What Could Go Wrong?

- Weak $Z \rightarrow T$ association: the instrument provides very little treatment variation.
- $Z \rightarrow Y$ direct effect: exclusion restriction is violated.
- $U \rightarrow Z$ association: independence fails.
- A statistically strong first stage does not validate exclusion.

Always draw the DAG and explain how the instrument was generated.

How IV Differs from Covariate Adjustment

Matching / IPW / AIPW

Require exchangeability after conditioning on measured X.

IV

Allows unmeasured $T - Y$ confounding, but needs an instrument satisfying new assumptions.

IV is not a universally safer estimator: its assumptions can be harder to defend.

Before Estimation: Four Questions

- What intervention and causal effect do we care about?
- Why should Z change treatment?
- Why is Z not connected to hidden causes of Y?
- Could Z influence Y through any pathway other than T?

Only then compute the IV estimator.

A Proposed IV: Physician Prescribing Preference

Clinical question

Does receiving Drug A rather than Drug B improve recovery?

Patients with worse underlying severity U may be more likely to receive A.

Candidate instrument

Z: physician prescribing preference

T: received Drug A

Y: recovery

U: hidden disease severity

Could physician quality or patient sorting violate independence or exclusion?

The Key Intuition: Instrument-Induced Variation

Observed T differences

Treatment selection reflects both Z and hidden factors U.

Z-induced T differences

Compare the change in outcomes associated with changes in Z.

IV isolates a particular source of variation in treatment.

The First Stage: Instrument → Treatment

$$\Delta T = E[T \mid Z = 1] - E[T \mid Z = 0]$$

- The first stage measures how strongly Z changes treatment.
- For binary T, ΔT is a difference in treatment probabilities.

A near-zero ΔT makes the Wald ratio unstable.

The Reduced Form: Instrument → Outcome

$$\Delta Y = E[Y \mid Z = 1] - E[Y \mid Z = 0]$$

- This is the observed difference in Y between instrument groups.
- Under IV assumptions, the effect is transmitted through treatment.

The reduced form alone is not necessarily the treatment effect.

The Wald Estimator: A Ratio of Differences

$$\text{IV effect} = \Delta Y \,/\, \Delta T$$

- Numerator: instrument–outcome association.
- Denominator: instrument–treatment association.
- Requires a nonzero first stage and credible IV assumptions.

With binary Z and T plus monotonicity, this identifies LATE.

Numerical Example: Encouragement to Attend Training

Instrument group	P(Training = 1)	Mean earnings
Z = 1: encouraged	0.70	$45,000
Z = 0: not encouraged	0.30	$41,000

First stage = 0.40; reduced form = $4,000; Wald = $10,000.

Calculate the Wald Estimate Step by Step

Wald = ($45,000 – $41,000) / (0.70 – 0.30)

- Reduced-form difference = $4,000.
- First-stage difference = 0.40.
- IV estimate = $10,000 per additional training participation.

Under LATE assumptions, interpret this as an effect for compliers.

Why Does Dividing Make Sense?

Step 1: shift Z

Encouragement increases training by 40 percentage points.

Step 2: observe Y

The encouraged group earns $4,000 more on average.

Step 3: rescale

$4,000 / 0.40 = $10,000 per induced participant.

The ratio translates the effect of encouragement into an effect of training.

Why Does the IV Ratio Work?

$$Y = \beta T + \gamma U + \varepsilon$$

If $Cov(Z, U) = 0$ and $Cov(Z, \varepsilon) = 0$, then:

$$Cov(Z, Y) = \beta \, Cov(Z, T)$$

$\beta = Cov(Z, Y) \,/\, Cov(Z, T)$, if $Cov(Z, T) \neq 0$.

This derivation assumes a constant structural treatment effect β.

Two-Stage Least Squares (2SLS)

Stage 1: predict treatment

Regress T on Z and baseline X to obtain fitted $\hat{T}$.

Stage 2: estimate outcome

Regress Y on $\hat{T}$ and baseline X, using proper IV inference.

Do not use ordinary second-stage lm() standard errors; use an IV routine.

2SLS with Baseline Covariates

$$T = \alpha_0 + \alpha_1 Z + \alpha^{\mathrm{T}} X + error$$

- Stage 1 obtains fitted treatment $\hat{T}$ from Z and X.
- Stage 2 uses instrument-generated variation in T to estimate its effect on Y.
- With constant linear effects and valid instruments, 2SLS targets that coefficient.

For heterogeneous effects, interpretation needs additional care.

Does IV Identify the Population ATE?

ATE

Average causal effect across everyone in the target population.

LATE

Average causal effect among people whose treatment is changed by the instrument.

A binary-instrument Wald ratio usually identifies LATE, not ATE.

Four Types of People (Binary Z and T)

Compliers

Take treatment if Z = 1, not if Z = 0.

Always-takers

Take treatment regardless of Z.

Never-takers

Never take treatment regardless of Z.

Defiers do the opposite of Z; monotonicity rules them out.

Monotonicity and the Compliance Types

$$T(1) \geq T(0) \text{ for every person}$$

- T(z): treatment received if the instrument were set to z.
- No one is discouraged into the opposite treatment direction.
- This is an assumption; data alone cannot establish it.

For a binary treatment and instrument, monotonicity excludes defiers.

Local Average Treatment Effect (LATE)

$$LATE = E[Y(1) - Y(0) \mid complier]$$

- Compliers are defined by their potential treatment responses to Z.
- LATE can differ from the ATE if effects vary across people.
- External validity depends on who the compliers are.

Always identify whose causal effect the instrument estimates.

ATE, ATT, and LATE: Not the Same Population

ATE

Everyone in the target population.

ATT

Those who actually received treatment.

LATE

Those whose treatment choice responds to Z.

Explain the estimand before interpreting the sign or magnitude.

Weak Instruments: Why First Stage Matters

Problem

If Z barely changes T, the denominator of the Wald estimator is near zero.

Consequence

Large uncertainty, unstable estimates, and unreliable usual approximations.

Practice

Inspect the first stage; consider weak-IV-robust intervals and tests.

Do not treat a first-stage F-statistic threshold as a universal guarantee.

Applied Example: Simulate a Confounded Setting

R workflow

```
set.seed(7900)
n <- 1500
Z <- rbinom(n, 1, 0.5)
U <- rnorm(n)
T <- 0.8 * Z + U + rnorm(n)
Y <- 3 * T + 2 * U + rnorm(n)
```

What to notice

Z is randomized.

U affects both treatment and outcome.

The true treatment coefficient is 3.

Why Ordinary Least Squares Is Biased

R workflow

```
d <- data.frame(Y, T, Z)
fit_ols <- lm(Y ~ T, data = d)
summary(fit_ols)
```

What to notice

U is missing from the fitted regression.

OLS mixes the treatment effect with confounding.

What direction should bias have?

Estimate the First Stage and Reduced Form

R workflow

```
first <- lm(T ~ Z, data = d)
reduced <- lm(Y ~ Z, data = d)
coef(first)["Z"]
coef(reduced)["Z"]
coef(reduced)["Z"] / coef(first)["Z"]
```

What to notice

The ratio reconstructs the Wald estimate in a just-identified linear setting.

First stage should be clearly nonzero.

Fit 2SLS with a Dedicated IV Routine

R workflow

```
# install.packages("AER")
library(AER)
fit_iv <- ivreg(Y ~ T | Z, data = d)
summary(fit_iv, diagnostics = TRUE)
confint(fit_iv)
```

What to notice

2SLS uses Z as the instrument for endogenous T.

Inspect IV coefficient, SE, confidence interval, and first-stage diagnostics.

OLS and IV: What Should the Simulation Show?

Population OLS coefficient

$$T = 0.8Z + U + \epsilon T$$
$$Y = 3T + 2U + \epsilon Y$$

$$Var(T) = 2.16, Cov(T, U) = 1$$

OLS slope $= 3 + 2/2.16 \approx 3.93$

Causal IV coefficient

True treatment coefficient: 3.00

Instrument Z is independent of U and outcome noise.

2SLS targets 3.00 as sample size grows.

For actual estimates run coef(fit_ols)["T"] and coef(fit_iv)["T"] in R.

Add Baseline Covariates Correctly

R workflow

```
# When baseline X is observed:
fit_iv_x <- ivreg(
  Y ~ T + X | Z + X,
  data = d_with_x
)
summary(fit_iv_x, diagnostics = TRUE)
```

What to notice

Put exogenous covariates X on both sides of the IV formula.

This is a separate template requiring a dataset with X.

How to Interpret the R Output

OLS estimate

May be biased because U predicts both T and Y.

IV estimate

May be consistent under valid instrument and structural assumptions.

Precision

IV often has larger standard errors; design credibility still matters.

In the simulation, the data-generating treatment coefficient is 3.

Method Comparison: What Changes?

PSM / IPW / AIPW

Measured conditional exchangeability, plus positivity and consistency.

Sensitivity analysis

Assume a particular class/strength of hidden bias to assess robustness.

Instrumental variables

Relevance, independence, exclusion; monotonicity for binary-IV LATE.

Different methods answer different identification problems.

Lecture 10: Take-Home Messages

Hidden confounding

Flexible estimators alone cannot repair failure of exchangeability.

Sensitivity analysis

Quantifies how results change under specified unobserved bias.

Instrumental variables

Uses a new identifying strategy, often estimating LATE.

Next lecture: Difference-in-Differences and policy evaluation.